

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Electronics and Electrical Communication Engineering
2. **Subject Name:** Algorithm Laboratory      **L-T-P:** 0-0-3      **Credits:** 2
3. **Pre-requisites:** None
4. **Syllabus and reference books:**

### Syllabus:

Queue, Stack and Heap, Karatsuba's Algorithm, Sorting, Fast Fourier Transform, Depth First Search, Bread First Search, Shortest Path - Dijkstra's and Bellman-Ford Algorithms, Strongly Connected Components, Directed Acyclic Graphs, Minimum Spanning Tree - Kruskal's and Prim's Algorithms, Set Cover, Huffman Encoding, Longest Increasing Subsequences, Edit Distance, Knapsack, Floyd-Warshall Algorithm, Travelling Salesman Problem, Simplex, Network Flows.

### Reference Books:

- 1) Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, Algorithms, McGraw-Hill Education, 2006.
- 2) Thomas Cormen, Charles Leiserson, Ronald L. Rivest and Clifford Stein, Introduction to Algorithms, MIT Press, 2009.
- 3) Jon Kleinberg and Éva Tardos, Algorithm Design, Pearson, 2013.
- 4) Udi Manber, Introduction to Algorithms: A Creative Approach, Addison-Wesley, 1989.

### 5. Lecture-wise break-up:

| Sl. No.               | Topic                                                                                                                                                                     | No. of lectures |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.                    | Queue, Stack and Heap Data Structures                                                                                                                                     | 1x3 = 3         |
| 2.                    | Divide and Conquer Algorithms: Karatsuba's Algorithm, Sorting, Fast Fourier Transform                                                                                     | 2x3 = 6         |
| 3.                    | Graph Algorithms: Depth First Search, Bread First Search, Shortest Path - Dijkstra's and Bellman-Ford Algorithms, Strongly Connected Components, Directed Acyclic Graphs. | 3x3 = 9         |
| 4.                    | Greedy Algorithms: Minimum Spanning Tree - Kruskal's and Prim's Algorithms, Set Cover, Huffman Encoding                                                                   | 2x3 = 6         |
| 5.                    | Dynamic Programming: Longest Increasing Subsequences, Edit Distance, Knapsack, Floyd-Warshall Algorithm, Travelling Salesman Problem.                                     | 4x3 = 12        |
| 6.                    | Linear Programming: Simplex, Network Flows.                                                                                                                               | 2x3 = 6         |
| Total number of hours |                                                                                                                                                                           | 42              |